

2023 updated version! With new modifications for the second round of tests.

Preregistration draft: Bitrex tests with multiple species (preregistration submitted to Open Science Framework on 5/19/23) <https://osf.io/br3xq> (not open access yet, under embargo)

2023 printable datasheet

Please **print** one of these datasheets for each test at the time that the test is set up in the lab. As the trial is being set up, record all relevant data on this paper sheet (e.g., spider ID, size, etc). Use the same paper datasheet to record data from the video. Keep hard copies of all datasheets in room 2007.

<https://docs.google.com/spreadsheets/d/1oDZ1GRIVl3xTMDJHdxLDwvVstPN3PBVj/edit#gid=440380700>

Link to DATA for 2023

This is where the data from the paper datasheets will be entered for analysis.

https://docs.google.com/spreadsheets/d/1QKrnsJZ9hjGGTGo794UxpAHVoCOH_Vdo4jH65Ope_w/edit#gid=0

Link to Kayla's older spreadsheet from 2022:

https://docs.google.com/spreadsheets/d/1uoyeyCbAmvQORKhSGljg43iQswMLBQD6eIYBQ_uJS34/edit#gid=0

Link to previous preregistration document for pilots in 2022:

<https://docs.google.com/document/d/1XEILUfi6y7q-GCKhhpv0lo6JV1zTx1aGLKLZbWzEqHU/edit>

~~Link to Google folder where videos and all documents related to this project can be stored.~~

~~There should be plenty of space here. If not, let Lisa know. Files take up space of the file "owner". If you don't have the space to store something lab-related, please transfer the file's ownership to Lisa (see instructions here:~~

~~<https://support.google.com/drive/answer/2494892?hl=en&co=GENIE.Platform%3DDesktop>.~~

~~All lab data is ultimately archived in Dryad and/or Open Science Framework, but we use Google Docs as a temporary place to collaborate.~~

~~<https://drive.google.com/drive/folders/1VtwxvvnJPyoBaw4-Pnw8ic2rEoiuWQU>~~

Title: Variation in responses to unpalatable prey across multiple jumping spider species

Authors: Lisa Taylor, Laurel Lietzenmayer, Aditi Persad (Plus others depending on interest - I'm happy to be inclusive on authorship here for all who are participating heavily in testing and are willing to help with writing, especially as we are adding in new species. This project is a big effort!)

General research question:

Do jumping spider species vary in their willingness to consume bitter-tasting prey?

Background:

Jumping spiders (family Salticidae) are a diverse group with more than 6400 described species to date (World Spider Catalog 2023). Recent work has revealed interesting differences in how two species respond to unpalatability and toxicity in prey, using quinine to make prey unpalatable (i.e., bitter) and springtails (*Fulsomia candida*) as toxic prey. In that study, we found that one species (*Habronattus trimaculatus*) was both less willing to eat unpalatable prey, and suffered more severe consequences from consuming toxic prey, compared to a second species (*Phidippus regius*) (Vickers, Heisey, and Taylor 2021). Yet, beyond that single study, little is known about how these two species differ in response to other types of unpalatable prey, and how patterns of responses vary across the rest of the family Salticidae.

This question is of particular interest as we try to make sense of the diverse visual capabilities, including variation in color vision, that have been documented across this family (e.g., Zurek et al. 2015, Outomuro et al. 2019). We have recently proposed that the evolution of enhanced color vision in some groups of jumping spiders (e.g., the genus *Habronattus*) may have been driven by a heightened susceptibility to prey toxins and the benefits these groups would gain from the ability to avoid aposematic prey that advertise their toxins with long-wavelength colors (Vickers, Heisey, and Taylor 2021). By contrast, species that are willing and able to consume toxic prey with little negative consequences (e.g., species in the genus *Phidippus*, see Powell and Taylor 2020, Vickers, Heisey, and Taylor 2021) may not face such strong selection pressures on their visual systems because avoiding aposematic prey is less important to their fitness.

Specific research question

How does aversion to unpalatable prey differ across Salticid species?

Hypotheses

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Our study addresses three main hypotheses:

- 1) Our first goal is to test the hypothesis that the findings from Vickers, Heisey, and Taylor (2021) reflect broader patterns of responses to prey unpalatability; specifically, that spiders in the genus *Habronattus* are *generally* more averse to unpalatability in prey compared with spiders in the genus *Phidippus*.
- 2) Our second goal is to expand beyond these two genera to test the hypothesis that other species vary in their responses to unpalatability. We don't have strong *a priori* reason to make directional predictions about how each of these species will compare to *Habronattus* and *Phidippus*; we simply have an expectation of variation across species because of the diverse natural histories of the species included in the study, and how

aspects of this diversity might affect the costs and benefits of avoiding or consuming toxic prey.

- 3) Our third goal is more practical: to test the hypothesis that the bittering agent Bitrex could be used reliably across jumping spider species as a tool in experiments that require training spiders to have particular prey aversions (e.g., Taylor et al. 2016, Winsor, Ihle, and Taylor 2020, and future work). Despite the expected variation across species, for this tool to be effective for any given species, we would need to ensure that that species indeed shows some aversion to Bitrex.

Design plan

Study type

Experiment

Blinding

Wherever possible, data will be collected by observers blind to treatment. Specifically, all data collected from video recordings (prey attacks, prey rejections) will be done by an observer blind to the treatment group of the prey item (i.e., whether the termite is DB treated or non-DB treated, see below). The treatment status of each prey item will be labeled with sticky notes during setup and these sticky notes will be removed just before the trial begins; as such, they will not be visible in the videos. At the setup of each trial, the position of the DB termite vs. non-DB termite in the arena will be assigned at random (one on the left and one on the right side of the arena). The positions of the termites will be recorded on a paper datasheet when the trial is set up and will not be visible to the observer scoring behavior from the videos.

Study design

1. To test the hypothesis that spiders in the genus *Habronattus* are generally more averse to unpalatability in prey compared with spiders in the genus *Phidippus*, we will present spiders from both genera with choices between unpalatable and palatable prey. Our hypothesis leads us to the directional prediction that *Habronattus* will show a stronger aversion to prey treated with the bittering agent denatonium benzoate (DB, marketed as 'Bitrex') compared with *Phidippus*. Anecdotal evidence suggests that these predicted patterns of aversion to unpalatability will apply to these two spider genera broadly; we will specifically use *Habronattus trimaculatus* and *Phidippus regius* in our study due to their ease of collection at known collecting sites.
2. To test the hypothesis that other species vary in their responses to unpalatability, we will perform the same tests as described above for testing Hypothesis 1, but will apply it to a broader range of species. As described above, the main difference between Hypothesis 1 and 2 is that for Hypothesis 2 we don't have strong *a priori* reason to make directional predictions about how each of these additional species will compare to *Habronattus* and *Phidippus*. While we might speculate on differences based on our knowledge of their

natural history, we don't yet have enough information to make strong *a priori* directional predictions about these particular species. In addition to *H. trimaculatus* and *P. regius*, we will also include the following additional species that are readily available at our current field sites: *Lyssomanes viridis*, *Anasaitis canosa*.

3. To test the practical hypothesis that Bitrex could be used reliably across jumping spider species as a tool in experiments that require training spiders to have particular prey aversions (e.g., Taylor et al. 2016, Winsor, Ihle, and Taylor 2020, and future work), we will assess whether all species included in our study have some degree of aversion to Bitrex. Despite the variation that we expect to see across species, for this tool to be effective broadly across the family, we would need to ensure that each species indeed shows some aversion to Bitrex (i.e., individuals of each species are more likely to reject DB-treated prey compared to non-DB treated prey).

Randomization

In each trial (i.e., each test with a single spider and two prey items), the position of DB termites vs. non-DB termites in the arena will be assigned at random (one on the left and one on the right side of the arena).

Sampling plan

Existing data

Registration prior to creation of data. As of the submission date of this preregistration, we have set up 55 trials across 3 of the 4 species to be tested (10 of 30 *Habronattus trimaculatus*, 21 of 30 *Anasaitis canosa*, 24 of 30 *Lyssomanes viridis*, and 0 of 30 *Phidippus regius*). Of these trials, 24 videos have been observed by blind observers, but no data have been collated, or analyzed. (Pilot data were collected in 2022 to help us establish our methods. We have made slight alterations to our methods since then and none of these 2022 pilot data will be included in any of the analyses proposed here.)

Data collection procedures

Spider collection and prey collection

Spiders of all species will be collected from known populations around Gainesville, FL. The spiders will be maintained following standard lab protocol and **will be tested within 7 days of collection** (to reduce variation associated with extended time in the lab). They will be fed approximately their own body size in crickets (*Gryllobates sigillatus*) and sprayed with water 3 times weekly (MWF) and housed in clear plastic boxes (10.16×10.16×12.86cm with a ~10cm

long artificial green plant affixed for enrichment (Carducci & Jakob 2000)). The boxes are opaque on one side to prevent spiders from seeing or interacting with one another. For each species, we will test juvenile (4mm in body length or greater) as well as adult female and male spiders (see target sample sizes for each below).

The termites used as prey in our experiments (*Reticulitermes flavipes*) will be collected from around Gainesville, FL and will be housed within cardboard and wooden logs in the laboratory.

Manipulation of prey palatability

To make prey unpalatable (i.e., to create “DB termites” for our tests), we will apply a 3% denatonium benzoate (DB) solution (Bitrex®) (Chemsavers, Bluefield, VA, USA) to termite prey following methods adapted from Winsor et al. (2020). Briefly, we will use a paintbrush to place a drop of DB solution onto the dorsal side of living termites (the drop should be about the size that comes out of a medicine dropper, and the excess will be wicked up by the filter paper). This application will be done in a filter paper lined petri dish and the termite will be allowed 5 minutes to dry. Because the application of DB might result in excess DB solution running onto the filter paper (and therefore excess DB on that filter paper), we will wait the 5 minutes for the DB to dry and then move each DB termite into a new/clean small petri dish (1cm x 3.5cm) lined with damp white filter paper. Sham-treated termites (i.e., “non-DB termites” for our tests) will be prepared by brushing a water solution on the termites and allowing them to dry and moving them into a new dish as described above. We will present termites to spiders within these dishes to ensure they cannot interact with one another (and therefore prevent the DB termites from contaminating the non-DB termites). The damp filter paper in the petri dishes will prevent the termites from desiccating during the 2-hour tests. Because DB is extremely bitter at even micro concentrations, care will be taken throughout this process to ensure that non-DB termites do not come into any contact with DB (or any tools, such as paintbrushes or dishes, that have touched DB). Different sets of tools will be used to handle DB and non-DB termites to avoid contamination.

Testing for aversions to Bitrex

We will maintain spiders on their standard diet until testing day. Because spiders are fed MWF, if their testing day falls on a MW or F, they will skip their feeding *that day*, but otherwise their feeding schedule will remain unchanged. (Note: That means that if a spider is tested on a Tuesday, they will still be fed on Monday. If they are tested on a Monday, their last feeding will have been Friday. The time since their last feeding will be recorded on the datasheet so that we can consider this in exploratory analyses. As an example, if they were fed on Monday and tested on Tuesday, the “time since their last feeding” to record on the datasheet will be 1 day). Spiders should have at least one feeding day in the lab before being tested.

In preparation for each test, we will prepare a clear plastic testing arena (a plastic box measuring 10.16×10.16×12.86cm). Onto the floor of this box, we will place two petri dishes (35mm), each lined with wet filter paper and no lid. Into one petri dish we will place a DB-treated

termite, and into the other we will place a non-DB termite. We will assign the position of these two types of termites (left or right side of the arena) at random for each test. We will then remove the test spider from their home box and place them into a third clear plastic petri dish (35mm diameter), enclosed with a clear lid. We will place the spider's petri dish between the two termite dishes for 2 minutes to allow the spiders to acclimate. During this acclimation period, the spiders are able to view the two termites moving in the vicinity, but the spiders will not be able to attack because they are confined within the dish. At the end of the 2-min acclimation period, the test spider will be released from the petri dish (by removing the clear lid) so that they can freely enter the termite dishes and attack the termites. We will record all interactions using overhead video cameras for 2 hours.

If the spider does not interact with either of the two termites during the 2-hour trial (i.e., if there are no attacks), they will be re-tested within the next two days (without being fed). If the spider does not interact with either of the two termites on their second trial, they will not be re-tested a third time.

From these videos, we will score the following behaviors for our confirmatory analyses: (1) whether or not the DB-treated termite was consumed (Y/N) during the two-hour trial, (2) whether or not the non-DB termite was consumed (Y/N) during the two-hour trial, (3) the number of times the DB-treated termite was attacked (or otherwise touched or bitten) and then rejected during the two-hour trial, (4) the number of times the non-DB termite was attacked (or otherwise touched or bitten) and then rejected during the two-hour trial. We will define "consumption" of a termite as the spider feeding on continuously for 2 minutes or more. We will define "rejection" of a termite as the spider attacking a termite and then dropping it (ceasing feeding) within 2 minutes of the attack.

For our exploratory analyses, we will record: (1) which of the two termites was attacked (or otherwise touched or bitten) first (DB-treated or untreated), (2) the approximate body size of the spider (total body length measured to the nearest cm), (3) the time (# of days) since the spider's last feeding.

We will test all spiders in an area of the laboratory that is supplemented with full spectrum lighting (SoLux PAR38 3500K 90W) and natural light from a large adjacent window. We will run all tests between 8am and 5pm, as this is the time of day when the spiders are most active (and therefore most willing to actively hunt termites during our trials). **Note: This means that the start time of the 2hr trials must fall between 8am and 3pm (so that all are completed by 5pm).**

Sample size

Our target sample size will be 30 individuals of each species (10 adult males, 10 adult females, and 10 juveniles) that interact with at least one of the two termites offered to them during the 2hr trials. Spiders that do not interact with either of the two termites will be re-tested once (as described above). If the spider does not interact with either of the two termites in their second

test, they will not be re-tested a third time and will not be included in our target sample size (meaning that we will continue to test spiders until we have 30 individuals of each species that have interacted with at least one termite in their tests).

Sample size rationale

Our target sample size of 30 is a balance between what is practical to collect/test and what our pilot data suggest will be needed to detect any effects. While we don't have *a priori* reason to expect differences in DB aversion among sexes and life stages, including adult males, adult females, and juveniles in our dataset will allow us to explore this.

Stopping rule

While aiming for our target listed above, we will stop once we hit 15 individuals in each sex/age category or if we are no longer able to collect spiders.

Variables

Manipulated variables

We will manipulate the palatability of termite prey using DB (as described above).

Measured variables

For our confirmatory analyses, we will record:

- (1) whether or not the DB termite was consumed (Y/N) during the two-hour trial
- (2) whether or not the non-DB termite was consumed (Y/N) during the two-hour trial
- (3) the number of times the DB-treated termite was attacked and then rejected during the two-hour trial
- (4) the number of times the non-DB termite was attacked and then rejected during the two-hour trial.

For our exploratory analyses, we will record:

- (1) which of the two termites was attacked first (DB-treated or untreated)
- (2) the approximate body size of the spider (total body length measured to the nearest cm)
- (3) the time (# of days) since the spider's last feeding

Indices

None

Analysis plan

Confirmatory analyses

(1) To test the hypothesis that spiders in the genus *Habronattus* are *generally* more averse to unpalatability in prey compared with spiders in the genus *Phidippus*:

- (a) We will test the directional prediction that *Phidippus* will be more likely to consume the DB termite compared with *Habronattus*. We will test this prediction using a 1-tailed X^2 test. The rationale for using a 1-tailed test is the *a priori* directional prediction.
- (b) We will calculate a DB aversion score (see below) for each individual and use a 1-tailed t-test (or nonparametric Mann-Whitney U) to ask if these values are greater for *Habronattus* compared with *Phidippus*. The rationale for using a 1-tailed test is the *a priori* directional prediction. The DB aversion score will be the number of DB rejections divided by the total number of rejections (within each 2hr trial). This value should reflect aversion to DB, but be independent of voraciousness/boldness. If the spider did not reject any termites in the trial (i.e., the denominator is 0), they will be given an aversion score of 0.

(2) To test the hypothesis that the other species included in this study vary in their responses to unpalatability:

- (a) We will use a likelihood ratio X^2 test to test the prediction that the species differ in their likelihood of consuming a DB termite.
- (b) We will use a non-parametric Kruskal-Wallis test to test the prediction that DB aversion scores vary by species.

(3) To test the practical hypothesis that DB could be used reliably across jumping spider species as a tool in experiments that require training spiders to have particular prey aversions (i.e., to test the prediction that individuals of each species are more likely to consume non-DB termites and more likely to reject DB-treated termites):

- (a) Within each species, we will use a 1-tailed X^2 test to test the directional prediction that the DB-treated termites are more likely to be consumed than the non-DB termites. The rationale for using a 1-tailed test is the *a priori* directional prediction.
- (b) Within each species, we will use a 1-tailed Wilcoxon signed rank test (the nonparametric equivalent of a paired t-test) to test the directional prediction that the DB-treated termites are rejected more often than the non-DB termites. The rationale for using a 1-tailed test is the *a priori* directional prediction.

Transformations

None needed. Data that do not meet parametric assumptions will be analyzed with non-parametric tests (as described).

Inference criteria

P-values ($\alpha = 0.05$) will be used to determine significance. For predictions that are explicitly directional (as described above) we will use one-tailed tests for hypothesis testing because this will provide the most power to test our hypotheses and because our data in these cases meet the two criteria recognized by Ruxton & Neuhäuser (2010) as appropriate for the use of one-tailed tests. Specifically, the *a priori* predictions are directional and a null result or significant result opposite the expectation will result in the same conclusion (i.e. rejection of our hypotheses). For predictions that are not-directional (as described above), we will use 2-tailed tests.

Exploratory analyses

(1) We will use likelihood ratio X^2 tests to explore whether the spiders' first attacks in a trial are directed equally between DB and non-DB termites (i.e., they do not differ from the expectation of 50/50). This will confirm previous findings (see Winsor et al 2020) suggesting that the spiders do not avoid the DB without first tasting it in the prey they attack. It is possible that they might detect DB from the filter paper in the dish with the DB termite, or smell it from the DB termite, but our previous work suggests that they do not.

(2) We will use nonparametric Kruskal-Wallis tests to explore whether the sexes or life stages (i.e., M, F, J) differ in their aversions to DB (using the DB aversion score as the dependent variable).

(3) Within each species, we will use a nominal logistic regression to explore whether the time from their last meal (i.e., # of days since last feeding) affects their likelihood of consuming a DB-treated termite. The link between these two variables is possible if hungrier spiders are less choosy, and therefore more willing to take prey that are distasteful.

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